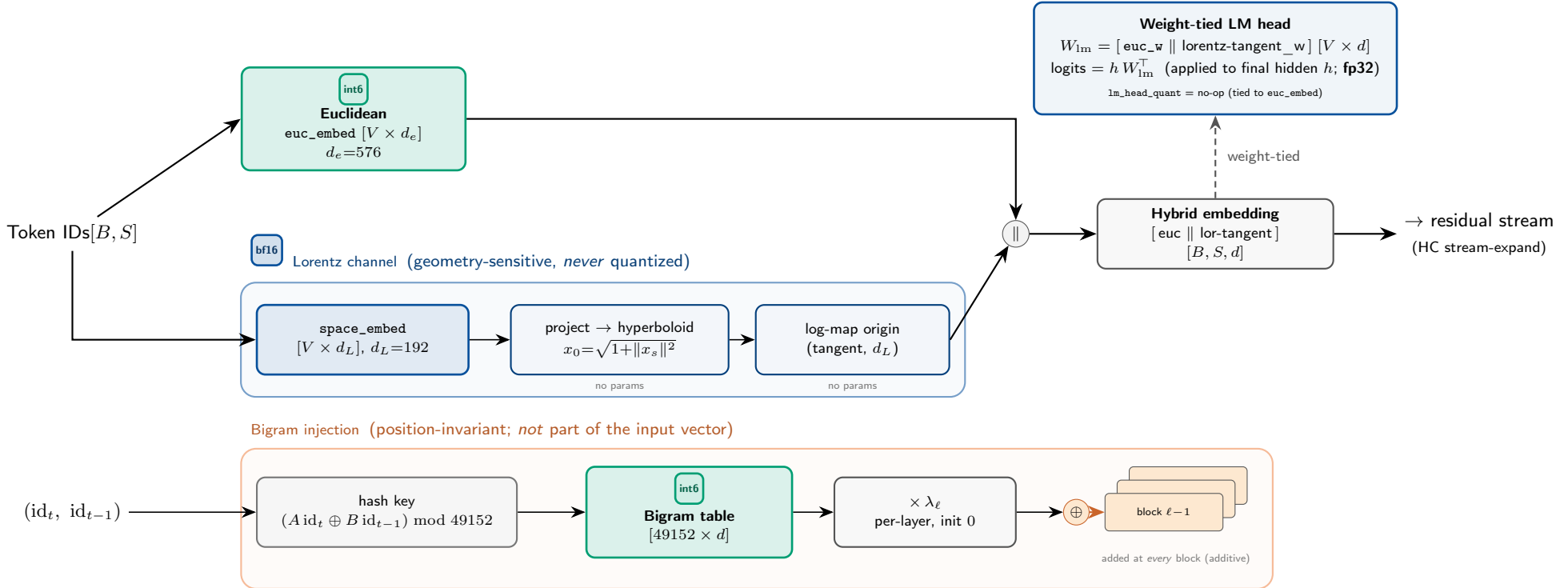


MORPH Embeddings — Hybrid Input + Bigram Injection

a hybrid Euclidean || Lorentz input vector (computed once) plus a position-invariant bigram signal injected per layer; the LM head is weight-tied to the Euclidean + Lorentz-tangent weights



Precision:

int6 euclidean + bigram tables (per-row int-N QAT)

bf16 Lorentz space_embed (geometry-sensitive, *always*) + all geometry ops + fp32 logits

Weight-tying: the LM head has *no* separate weight — W_{lm} is $[\text{euc_w} \parallel \text{lorentz-tangent_w}]$, so quantizing euc_embed already moves the head; lm_head_quant is a no-op at the weight level. Logits are cast **fp32** before the chunked softmax (fused_ce.py).

Notation: V vocab, d model dim, $d_e + d_L = d$ (lorentz_fraction= 0.25 $\Rightarrow$ 576+192 at $d=768$), λ_ℓ per-layer bigram gate (init 0, learned). Default embed_quant="int6" quantizes Euclidean and bigram rows; Lorentz stays bf16.